

5(a)	force per unit charge	B1
	(force on) positive charge	B1
5(b)(i)	field changes <u>direction</u> (between A and B)/field is zero at a point (between A and B)	M1
	so charges have same sign	A1
5(b)(ii)	Any one from: • field is (also) influenced by charge B • charge A is not isolated/is not the only charge present • field is due to two/both charges • field is the resultant of two fields	B1
5(b)(iii)	$E = Q / (4\pi\epsilon_0 x^2)$	C1
	at $x = 10 \text{ cm}$, $E_A = E_B$	C1
	$Q_A / 10^2 = Q_B / 5^2$ $Q_A / Q_B = 4.0$	A1